

Statistical analysis of the Ames *Salmonella*/microsome test

(Ames test/mutagenesis/negative binomial model/statistical modeling/toxicity)

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ABSTRACT A family of statistical models for analysis of Ames *Salmonella*/microsome test data is constructed that considers mutation and toxicity as competing risks and allows hyper-Poisson variability. These models have a parameter that can be employed as a mutagenic index because it approximates the slope at zero dose of a dose–response curve adjusted for toxicity. A second parameter quantifies plate-to-plate variability and lends itself to the study of factors affecting internal test reproducibility. The detection of aberrant plate counts is also addressed. This methodology is illustrated with data from a *Salmonella* test.

The Ames *Salmonella*/microsome test (1) currently holds a preeminent position among the various tests available to the genetic toxicologist for investigation of a chemical's mutagenicity. Indeed, in many quarters, a positive *Salmonella* test result is presumptive evidence for carcinogenicity. Although various authors (2–5) have proposed analyses of data from this test, each treatment has been deficient in some way: a Poisson distribution model for the sampling variability of the data is assumed, the possibility of toxicity is disregarded or treated in an ad hoc manner, or the multigenerational aspect of the test is ignored.

There is substantial empirical evidence that the Poisson model, nearly universal in its adoption as the sampling distribution of revertants per plate from a *Salmonella* test, lacks the flexibility to adequately describe the variability in a set of plate counts. This can lead to false conclusions concerning the outcome of the assay. An alternative sampling distribution is proposed that incorporates a measure of experimental reproducibility able to reflect dependence of internal variability among a set of replicate plates on the specific laboratory, technician, strain, or any other factor influencing the results from a single test.

Toxicity denotes any genetic or cellular phenomenon precluding reversion to histidine prototrophy or expression of the his⁺ phenotype. Dose–response curves from a test frequently exhibit departures from monotonicity because of toxicity. A biologically motivated family of mathematical models is constructed to relate plate counts, on average, to their respective doses of the test chemical. This family considers both mutation and toxicity to be dose-dependent competing risks that confront each *Salmonella* bacterium. These models are limited in their abilities to resolve ambiguities caused by a limited understanding of the behavior of a chemical once it is placed on a test plate.

A CRITIQUE OF THE POISSON MODEL

The Poisson distribution has been employed extensively in statistical modeling of microbial data. Its applicability to *Salmonella* test data appears to be a consequence of the following set of assumptions. (i) All microbes, whether on the same plate or not, behave in a stochastic manner independently of each other.

(ii) Each microbe placed on a particular plate (and its lineage) experiences the same environment as any other microbe (and its lineage) plated on the same plate; this environment, denoted by ξ , is a function of the test compound and its dose, the amount of histidine present, and sundry other factors. (iii) The targeted number of microbes placed on a plate, denoted by N_0 , is large—e.g., 10^8 . (iv) The probability, $P(\xi)$, that any single plated microbe, through its descendants, gives rise to a visible revertant colony is small, say 10^{-5} to 10^{-9} , and is not altered by the number of microbes plated.

These four assumptions imply that the number of revertant colonies observed on a plate with environment ξ , denoted by X_ξ , has approximately a Poisson distribution with mean parameter $N_0P(\xi)$ (6)—i.e., for any nonnegative integer k , the probability that X_ξ equals k is given by

$$\text{pr}\{X_\xi = k\} = [N_0P(\xi)]^k \exp[-N_0P(\xi)]/k!, \quad [1]$$

in which $\exp[t] = e^t$.

For a set of r replicate plates with observed counts $X_{\xi 1}, X_{\xi 2}, \dots, X_{\xi r}$, it is commonly assumed (e.g., refs. 2, 3, and 5) that these r counts are adequately described as a random sample from Eq. 1. This does not follow solely from the four assumptions above but rests on an additional assumption: the r replicated plates share sufficiently similar environments and plated numbers of microbes, so that the realized mean parameters are essentially constant across replicates. If one accepts assumptions i–iv, the validity of the additional assumption can be evaluated empirically through a standard test of whether observed counts $X_{\xi 1}, X_{\xi 2}, \dots, X_{\xi r}$ behave as a random sample from Eq. 1.

To investigate this, four laboratories, without being told the intended purpose, were requested to produce 20 replicate plate counts under solvent control conditions with *Salmonella* strain TA100. This strain was selected because it is often cited as the most sensitive of the *Salmonella* strains. The number of plates was chosen to be that round number nearest the total number of plates involved in a single strain experiment consisting of a zero dose or solvent control and five or six chemical doses, with three replicate plates per dose; this experimental design was a recent mutagenicity conference recommendation (7). The resultant counts, ordered by magnitude, together with various summary statistics, are in Table 1.

The statistic T_ξ evaluated for each data set is

$$T_\xi = \sum_{i=1}^r (X_{\xi i} - \bar{X}_\xi)^2 / \bar{X}_\xi, \quad [2]$$

in which $\bar{X}_\xi = (1/r)\sum X_{\xi i}$. The statistic T_ξ is proportional to the ratio of sample variance to sample mean and is the usual statistic used to test whether a random sample of size r exhibits behavior consistent with Eq. 1. If the underlying distribution is sufficiently near a Poisson distribution, then the distribution of T_ξ is well-approximated by a χ^2 distribution with $(r - 1)$ degrees

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Table 1. Twenty replicated solvent control plate counts with *Salmonella* TA100 from four laboratories

Laboratory*				
1	2	3	4H†	4D†
81, 86	63, 64	74, 77	168, 171	132, 134
93, 97	65, 68	81, 85	174, 175	144, 145
99, 104	69, 70	87, 88	185, 189	145, 145
105, 110	72, 73	89, 90	190, 191	157, 158
112, 112	75, 80	93, 97	195, 197	158, 161
113, 114	82, 83	98, 99	198, 198	162, 164
115, 117	83, 84	102, 103	203, 205	166, 169
118, 122	84, 85	105, 108	205, 207	174, 177
131, 135	90, 91	108, 110	210, 214	201, 208
155, 183	‡, ‡	111, 124	216, 218	208, 219
Mean				
115.10	76.72	96.45	195.45	166.35
Variance				
540.62	81.15	163.10	226.58	631.29
T_ξ				
89.24	17.98	32.13	22.03	72.10
$\hat{c} (\times 10^{-2})$				
2.7	0.0	0.6	0.1	1.5
SE ($\hat{c}$) ($\times 10^{-2}$)				
1.1	0.4	0.5	0.2	0.7

* Revertant colonies per plate listed in ascending order within each laboratory and paired solely for conciseness of presentation.

† Laboratory 4 conducted this experiment twice, once with water (4H) and once with dimethylsulfoxide (4D) as the solvent. The same individual did the two tests but on different days. All other laboratories used water.

‡ Contaminated plate, not counted.

of freedom (8). For 20 replicates, this would imply that T_ξ exceeds 30.1 once in 20, 36.2 once in 100, and 43.8 once in 1000. Table 1 clearly demonstrates that these data are not Poisson distributed in all laboratories at all times. Zeiger *et al.* (9) and Vollmar (cited in ref. 10) have previously reported this finding. Although Vollmar stated that a Poisson distribution can be assumed up to about 150 revertants per plate, the data in Table 1 contradict this conclusion.

The main practical implication of Table 1 is that the variability of any statistic used to quantify or judge the evidence for mutagenicity may be substantially underestimated if one disregards the possibility of hyper-Poisson variability. To the extent that the results in Table 1 typify laboratory functioning, an underestimate of the variance by a factor of 2–4 is not rare. This, in turn, would elevate the real false-positive rate, and possibly even the real false-negative rate, and would yield confidence limits for parameters of interest whose coverage rates are overstated. The degree of elevation of the two error rates is dependent on the statistical procedure used. A later example will demonstrate how variance underestimation affects model parameter estimates. For present illustrative purposes, consider a one-tailed test for mutagenicity based on a statistic that is approximately normally distributed with mean 0 and variance $\rho\sigma^2$ under an assumption of no mutagenicity. If this statistic is treated as if its variance is σ^2 , then the real false-positive rates for this test when nominally 5% and 1% are respectively 12% and 5% if $\rho = 2$ and 21% and 12% if $\rho = 4$. The possible presence of hyper-Poisson variability underscores the need for replicate plates; only by this means can one properly quantify internal

plate-to-plate variability and, thereby, validly weigh the evidence for mutagenicity.

AN ALTERNATIVE SAMPLING DISTRIBUTION

It is our conjecture that laboratories occasionally exhibiting hyper-Poisson variability in a particular experiment are unable that day to maintain near constancy across replicates of the plate environment because of variability in pipetting the microbes, the agar, the S9 liver homogenate, and the test compound. For each plate, the Poisson mean would then be a stochastic quantity, having its own sampling distribution G . It follows that a set of replicate plate counts are a random sample from a mixture of Poissons represented by

$$\text{pr}\{X_\xi = k\} = (k!)^{-1} \int_0^\infty \tau^k e^{-\tau} dG(\tau). \quad [3]$$

The distribution G must be that of a nonnegative random variable and is probably unimodal, the targeted mean determining the mode if no systematic bias is involved. The distribution G that affords greatest mathematical tractability is that associated with a gamma density:

$$g(t; \mu, c) = [(\mu c)^{c-1} \Gamma(c)^{-1}]^{-1} t^{c-1} \exp[-t/\mu c], \quad [4]$$

in which $\mu > 0$, $c > 0$. With this choice, Eq. 3 reduces to a negative binomial distribution:

$$\text{pr}\{X_\xi = k | \mu, c\} = \binom{k + c^{-1} - 1}{k} [\mu/(\mu + c^{-1})]^k \times [c^{-1}/(\mu + c^{-1})]^{c^{-1}}, \quad [5]$$

in which k is a nonnegative integer. This distribution has mean μ and variance $\mu(1 + c\mu)$, which is greater than the mean. The next section discusses modeling μ as a function of applied dose.

The negative binomial can be evaluated at $c = 0$ through a limit argument to obtain the Poisson distribution; hence, large values of c are evidence for departures from Poisson sampling behavior. Aside from unimodality and tractability, the gamma density assumption rests on the historical role of the resultant negative binomial distribution as an important alternative to the Poisson model in microbiology; its use for this purpose dates back at least to Gossett [Student] (11), who was concerned with counting yeast cells using a haemocytometer.

The parameter c reflects the intrinsic precision of an experimenter in producing replicated plate environments on a given day, despite the varying test chemical dose. Perfect reproducibility of replicated plate environments is represented by $c = 0$. Variability in replicated plate environments is represented by $c > 0$: the greater the variability, the larger the value of c . Table 1 includes the maximum likelihood estimate $\hat{c}$ of c and its estimated standard error (12) for each of the five data sets.

Because the variance of a negative binomial random variable is a function of both μ and c , one is unable to estimate the standard error of an observation y solely from its model-fitted or predicted value $\hat{y}$, something that is obtainable for Poisson distributed data simply as $\hat{y}^{1/2}$. If c is nearly constant within an experiment, then the standard error for y can be estimated by $[\hat{y}(1 + \hat{c}\hat{y})]^{1/2}$. Thus, an individual observation may be checked empirically for aberrance by computing its absolute standardized residual:

$$|y - \hat{y}| / [\hat{y}(1 + \hat{c}\hat{y})]^{1/2}. \quad [6]$$

An illustration of this diagnostic check occurs later in the text.

MODELING DOSE RESPONSE

The lack of universal Poisson sampling behavior for *Salmonella* test data is one complication of the statistical analysis of such experiments. Additional complexity results from the frequently nonmonotonic relationship of revertants per plate to test chemical dose, a relationship that increases to a peak and then decreases, sometimes to a value below the solvent control values and occasionally to zero. This departure from a nondecreasing response is commonly ascribed to microbial toxicity, both genetic and cellular, caused by the test chemical. Were the currently recommended *Salmonella* test procedure to include a parallel assay that accurately measured the number of microbes succumbing to toxicity on the treated plates, one would then be able to modify statistical analyses to adjust for the effects of toxicity. However, this dual experiment is not common practice because of cost and convenience.

An effective way to disentangle the competing risks of mutation and toxicity in order to draw inferences regarding the former is to model μ as $N_0 P_D$, where N_0 is the average number of microbes placed on a plate, and P_D is the probability that a single plated microbe yields a revertant colony when exposed to dose D of a test chemical. Haynes and Eckardt (13) have recently come to a similar conclusion and have proposed models for P_D incorporating toxicity; their models are essentially of the form:

$$P_D = \{1 - \exp[-H_M(D)]\} \exp[-H_T(D)], \quad [7]$$

in which $H_M(D)$ and $H_T(D)$ are typically low-order polynomials and represent respectively the average number of mutagenic and toxic "hits" per microbe when dose D is applied to a plate. The simplest and most common case of Eq. 7 is the "single hit" or linear model, where

$$P_D = \{1 - \exp[-(\alpha + \beta D)]\} \exp[-\gamma D], \quad [8]$$

$$\alpha > 0, \beta \geq 0, \gamma \geq 0.$$

Models of the form in Eq. 7 have been investigated and discussed extensively in the field of radiation mutagenesis but may be inappropriate for chemical mutagenesis. Radiation is an instantaneous process; in response to a single burst, microbes either succumb to toxicity, mutate with regard to a specified genetic trait, or survive unchanged with regard to that trait. If a plated microbe escapes the toxic effects of radiation but mutates with respect to the specified trait, it is essentially assured of giving rise to a mutant colony. For this process, the plated microbes are the only ones to run the twin risks of mutation and toxicity. In the *Salmonella* plate assay, however, various alternatives are possible depending upon the test chemical. If it loses its mutagenic and toxic capabilities rapidly, say in a single microbial generation, then the process closely resembles that of single-burst radiation; on the other hand, if it is active for an extended period of time, then the plated microbes and their descendants are at risk of both mutation and toxicity. Because the activity of a chemical on a plate is not generally known, this uncertainty argues for broadening the class of models in Eq. 7. To this end, in addition to assumptions *i-iv* and the negative binomial sampling assumption, the following are assumed. (v) The probability of plating preexisting mutants or of experiencing forward mutation from prototrophy to auxotrophy is negligible. (vi) Expression occurs within the first generation following mutagenesis. (vii) Sufficient histidine is placed on each plate to permit auxotrophic growth through k generations, where k is typically three to six. (viii) Mutation and toxicity are stochastically independent processes. A microbe on a plate with dose D of the test chemical has probability $p_M(D)$ of mutating from

his⁻ to his⁺ if it is a member of the first m generations after plating, probability of $p_M(0)$ of mutating if it is a member of the next $k-m$ generations, and a zero probability thereafter. Both auxotrophs and prototrophs have probability $p_T(D)$ of succumbing to toxicity if they are members of the first t generations since plating and zero probability thereafter.

The step-function behavior assumed in *viii* is intended as a parsimonious approximation to probabilities that vary in a complex fashion with generation time. Because measurable mutation effectively ceases with histidine depletion (if not before) and because measurable toxicity (if present) is not necessarily so restricted, it has been assumed in *viii* that $m \leq k$, and $m \leq t \leq \infty$. In fact, for the single-hit assumptions subsequently adopted, setting $m = k$ introduces negligible error in the parameter estimates. Here $t = \infty$ is intended to indicate long-lasting toxicity (e.g., the entire test time of 48 or 72 hr). The assumptions above permit the following modeling.

Let $P_D(m, t)$ denote the probability that a plated microbe exposed to dose D of the test chemical gives rise to a revertant colony when mutation is present for m generations and toxicity for t generations. Similarly define $Q_D(s)$ to be the probability that any microbe and its descendants exposed to dose D become extinct because of toxicity within s generations, counting the plated microbe as generation one. Then upon consideration of the possible first generation events, one obtains the following recursion:

$$P_D(m, t) = p_M(D)[1 - p_T(D)][1 - Q_D^2(t-1)] + [1 - p_M(D)] \times [1 - p_T(D)][1 - [1 - P_D(m-1, t-1)]^2]. \quad [9]$$

The quantity $Q_D(s)$ also satisfies a recursion. If $1 \leq s \leq t$, then

$$Q_D(s) = p_T(D) + [1 - p_T(D)]Q_D^2(s-1);$$

otherwise $Q_D(s) = 0$ if $s = 0$, or $Q_D(s) = Q_D(t)$ if $s > t$. Useful values of $Q_D(s)$ are $Q_D(1) = p_T(D)$ for all $t \geq 1$,

$$Q_D(2) = p_T(D) + [1 - p_T(D)]p_T^2(D) \quad \text{for all } t \geq 2,$$

and

$$Q_D(\infty) = \min\{1, p_T(D)/[1 - p_T(D)]\} \quad \text{for } t = \infty.$$

Because $P_D(r, v)$ is very small for all $r \leq m$ and $v \leq t$, Eq. 9 may be approximated by

$$P_D(m, t) \doteq p_M(D)[1 - p_T(D)][1 - Q_D^2(t-1)] + 2[1 - p_M(D)][1 - p_T(D)]P_D(m-1, t-1). \quad [10]$$

If $m = k$, Eq. 10 admits the following solution:

$$P_D(m, t) = p_M(D)[1 - p_T(D)] \left\{ \sum_{j=0}^{m-1} (2[1 - p_M(D)] \times [1 - p_T(D)]^j \times [1 - Q_D^2(t-1-j)]) \right\}. \quad [11]$$

Four models are boundary members of the family of models in Eq. 11 subject to $m \leq \min(k, t)$; because they straddle the family, these models give good evidence of the varied behaviors possible in Eq. 11:

I. For $m = k = 1 = t$: $P_D(1, 1) = p_M(D)[1 - p_T(D)]$.

This is Eq. 7 with $p_M(D) = 1 - \exp[-H_M(D)]$ and $p_T(D) = 1 - \exp[-H_T(D)]$.

II. For $m = k = 1, t = \infty$:

$$P_D(1, \infty) = p_M(D)[1 - p_T(D)][1 - Q_D^2(\infty)] = p_M(D) \frac{[1 - 2p_T(D)]_+}{[1 - p_T(D)]},$$

i.e., mutation for one generation but long-lasting toxicity; $[x]_+ = \max(0, x)$.

- III. For $m = k = t > 1$: $P_D(k, k)$ in Eq. 11 admits no further simplification; both mutagenicity and toxicity are histidine-limited processes.
- IV. For $m = k > 1$, $t = \infty$: $P_D(k, \infty) = P_D(1, \infty) \sum_{j=0}^{k-1} (2[1 - p_M(D)][1 - p_T(D)])^j$. Here mutation is histidine-limited, but toxicity, if present, is long lasting.

For the remainder of the discussion, simplifying single-hit assumptions for mutation and toxicity are adopted, and m is set to k . These plus earlier assumptions lead to the following: $p_M(D) = 1 - \exp[-(\alpha + \beta D)]$, $\alpha > 0$, $\beta \geq 0$, and $p_T(D) = 1 - \exp[-\gamma D]$, $\gamma \geq 0$. The key parameter in this case is β , because the chemical under study is mutagenic if and only if $\beta > 0$; it is the only index of mutagenicity to date that both derives from mechanistic modeling of the Ames test and is adjusted for toxicity.

One distinction among the models is immediately apparent. The duration of toxicity on a plate can have a marked impact on the form of P_D and, hence, μ . For example, model I describes survival by the exponential decay function $\exp[-\gamma D]$, whereas model II depicts it as the "shoulder-shaped" function $[2 - \exp(\gamma D)]_+$, reaching zero at a finite dose $D = \gamma^{-1} \ln 2$ for $\gamma > 0$. The visible effects of toxicity are usually apparent only at high doses. If the positive doses are selected to be equispaced on a logarithmic scale, then there may be little information regarding the form of the survival function. As the example in the next section illustrates, models I and II may fit the data equally well, as judged by the evaluated logarithmic likelihoods, yet the differences in their estimates of the parameter β can frequently exceed that which is explainable by sampling variability. Of greater concern are differences in estimates of β that models II and IV can exhibit because the duration of mutagenicity on a plate is unknown. The example below illustrates this point.

APPLICATION AND IMPLICATIONS OF THE MODELS

Quinoline was studied by W. Speck (unpublished) as part of the Environmental Mutagenesis Test Development Program at the National Institute of Environmental Health Sciences. The data in Table 2 were obtained from a test with *Salmonella* strain TA98 and induced rat liver homogenate S9. Dimethyl sulfoxide was the solvent control, and each dose was replicated three times. Models I–IV, together with the single-hit and negative binomial assumptions, were fitted to these data by maximum likelihood techniques (14). Parameter estimates of α , β , γ , and c and their estimated standard errors are given in Table 3. A fifth model (V) that assumes $m = k = 6$ and $t = \infty$ is included as a further check on model sensitivity.

The fitted values of these five models (Table 4) show substantial agreement in fitting the data despite the differences in

Table 3. Parameter estimates from negative binomial single-hit models I–V for quinoline data

	Model				
	I ($m = 1$, $t = 1$)	II ($m = 1$, $t = \infty$)	III ($m = 3$, $t = 3$)	IV ($m = 3$, $t = \infty$)	V ($m = 6$, $t = \infty$)
$\hat{\alpha}(\times 10^{-8})$	20.3	22.0	2.99	3.04	0.325
$SE(\hat{\alpha})(\times 10^{-9})$	26.1	26.4	3.73	3.74	0.413
$\hat{\beta}(\times 10^{-10})$	21.5	10.7	2.38	2.14	0.318
$SE(\hat{\beta})(\times 10^{-11})$	62.7	32.4	8.17	5.90	0.898
$\hat{\gamma}(\times 10^{-4})$	21.2	5.72	5.71	5.14	3.79
$SE(\hat{\gamma})(\times 10^{-5})$	35.3	4.01	15.2	5.07	5.61
$\hat{c}(\times 10^{-2})$	5.36	6.53	5.45	6.02	5.49
$SE(\hat{c})(\times 10^{-2})$	2.98	3.37	3.01	3.20	3.02

model form and parameter estimates. The five logarithmic likelihoods obtained were -63.20 , -64.36 , -63.73 , -63.86 , and -63.34 , respectively. Estimates of the mutagenic index β vary from 2.15×10^{-9} to 3.18×10^{-11} . Models I and II differ by a factor of 2 in this estimate, which is solely attributable to differences in their modeling of toxicity. The wide variation in estimates of β among models II, IV, and V, all of which assume long-lasting toxicity, can be elucidated. If toxicity is essentially negligible, then because $p_M(D)$ is generally very small one can closely approximate Eq. 11 by:

$$P_D(m, t) = (2^m - 1)p_M(D) \\ = (2^m - 1)\{1 - \exp[-(\alpha + \beta D)]\}. \quad [12]$$

Consequently, it is virtually impossible to distinguish empirically between the models associated with $P_D(m, t)$ and $P_D(1, t)$, since their fits to the data will be nearly the same. However, their parameter estimates for α and β , denoted by $(\hat{\alpha}^{(m)}, \hat{\beta}^{(m)})$ and $(\hat{\alpha}^{(1)}, \hat{\beta}^{(1)})$, respectively, will differ by nearly a factor of $(2^m - 1)$ —i.e., $\hat{\alpha}^{(1)} \doteq (2^m - 1)\hat{\alpha}^{(m)}$ and $\hat{\beta}^{(1)} \doteq (2^m - 1)\hat{\beta}^{(m)}$. The respective standard errors of these estimates will differ by the same factor. The estimates of β in Table 3 exhibit comparable multipliers in the presence of mild toxicity and moderate sampling variability—e.g., $\hat{\beta}^{II} \doteq 5 \hat{\beta}^{IV}$ and $\hat{\beta}^{II} \doteq 34 \hat{\beta}^V$.

Therefore, in the presence of negligible toxicity and sufficient histidine for more than one generation, if it is assumed that mutagenic activity persists for m^* generations but in reality it persists for m generations, then the estimate $\hat{\beta}$ actually estimates approximately $(2^m - 1)/(2^{m^*} - 1)$ times the real β . If sufficient histidine for four or five generations is supplied to each plate, then the uncertainty regarding the mutagenic index β can be nearly an order of magnitude. This is far greater than the uncertainty evidenced by the estimated standard errors in Table 3, which must be interpreted as uncertainties conditional on the validity of the corresponding models. This uncertainty will affect any quantitative analysis that attempts to measure mutagenic "potency" because the number of microbes at risk of mu-

Table 2. Quinoline data (TA98, rat liver homogenate S9)

Revertant colonies with increasing quinoline*					
0	10	33	100	333	1000
15	16	16	27	33	20
21	18	26	41	38	27
29	21	33	60	41	42

* Revertant colonies per plate are listed in ascending order within each dose group (μg of quinoline per plate). Three plates were tested per dosage.

Table 4. Fitted values for quinoline data

Quinoline, μg per plate	Model					Averaged plate counts
	I	II	III	IV	V	
0	20.3	22.0	20.9	21.3	20.5	21.7
10	22.0	22.9	22.3	22.5	22.1	18.3
33	25.6	25.0	25.2	25.1	25.4	25.0
100	33.9	30.8	32.6	31.9	33.4	42.7
333	45.5	45.5	46.4	45.7	45.5	37.3
1000	28.4	29.4	28.9	29.1	28.6	29.7

tation is not known. Nevertheless, it is conceivable that for certain purposes one can accept an order of magnitude of uncertainty in estimating β . Alternatively, one can theoretically reduce this uncertainty by supplying only enough histidine for a small number of generations. For identifying mutagens, as opposed to the potentially more useful quantification of the strength of mutagenicity, the natural one-tailed statistical test that treats $S = \hat{\beta}/SE_{\hat{\beta}}$ as a standard normal random variable in the absence of mutagenicity is much less affected by the uncertainty of the number of microbes at risk of mutation. The values of S for the five models considered here are 3.43, 3.30, 2.91, 3.62, and 3.54, all of which indicate that quinoline is mutagenic.

The distinction between the negative binomial and Poisson sampling models can be inferred from Table 3 and the following values of $\hat{\beta} \pm SE_{\hat{\beta}}$ obtained for models I-V, respectively, under Poisson distribution sampling: $(2.04 \pm 0.363) \times 10^{-9}$, $(9.67 \pm 1.69) \times 10^{-10}$, $(2.24 \pm 0.466) \times 10^{-10}$, $(1.97 \pm 0.321) \times 10^{-10}$, and $(2.99 \pm 0.512) \times 10^{-11}$. The estimates of β for the quinoline data under the Poisson and negative binomial models exhibit small differences for any one of the five models considered, but their respective standard errors under the negative binomial model are approximately 1.75 times larger than under the Poisson model. This is intuitively reasonable because the two sampling models as used here differ not in expected values but in variances.

As mentioned earlier, one can use Eq. 6 to check for evidence of aberrant observations that might unduly influence the estimated β . For the data on quinoline, models I-V all point to the observed 60 revertant colonies at a dose of 100 μg per plate as the candidate for designation as a rogue observation. The values of Eq. 6 for the reading of 60 under each of the five negative binomial models are 2.67, 3.03, 2.88, 2.91, and 2.73, respectively; there is marginal-to-moderate evidence for aberrance in the reading of 60, depending upon the model employed in the analysis. The implication of this finding can be investigated by refitting each model to the quinoline data set with the observation of 60 deleted. The new values of $\hat{\beta}$ plus or minus their standard errors for the five models are $(1.55 \pm 0.447) \times 10^{-9}$, $(8.58 \pm 2.27) \times 10^{-10}$, $(1.78 \pm 0.584) \times 10^{-10}$, $(1.67 \pm 0.427) \times 10^{-10}$, and $(2.35 \pm 0.653) \times 10^{-11}$, respectively, which does not alter the conclusion that quinoline is mutagenic. The major impact of omitting the 60-revertants reading is to reduce $\hat{c}$, which reflects lack of fit and lack of reproducibility, both of which are reduced when the most discrepant observation is omitted from the modeling. The values of $\hat{c}$ upon deletion of the observation of 60 are 0.0212, 0.0249, 0.0215, 0.0231, and

0.0215, respectively. Each model still leads to the conclusion that the sampling distribution is not Poisson.

DISCUSSION

The models constructed and discussed here are approximations to the biochemistry and genetics of the Ames *Salmonella* test that ignore possible complexities of pharmacokinetics, threshold phenomena, and expression delay. This family of models provides a methodology for analysis of mutagenicity data that does not presuppose or exclude either Poisson variability or toxicity or rely on subjective judgments of density of the background lawn to determine whether toxicity is present. The parameters in these models serve different purposes. The parameter α can be employed to check the credibility of solvent control data vis-a-vis historical values. The parameter β represents a mutagenic index that estimates the instantaneous slope at zero dose of the observed dose response adjusted for toxicity. Finally, the parameter c serves to quantify plate-to-plate variability within a test and can be used as a means to implement quality control for a test protocol that is rapidly approaching a level of mass production in many laboratories.

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1. Ames, B. N., McCann, J. & Yamasaki, E. (1975) *Mutat. Res.* **31**, 347-364.
2. Katz, A. J. (1979) *Mutat. Res.* **64**, 61-77.
3. Weinstein, D. & Lewinson, T. M. (1978) *Mutat. Res.* **51**, 433-434.
4. Venitt, S. & Crofton-Sleigh, C. (1979) *Mutat. Res.* **68**, 107-116.
5. Stead, A. G., Hasselblad, V., Creason, J. P. & Claxton, L. (1981) *Mutat. Res.* **85**, 13-27.
6. Feller, W. (1967) *An Introduction to Probability Theory and Its Applications* (Wiley, New York), 3rd. Ed., Vol. 1.
7. de Serres, F. J. & Shelby, M. D. (1979) *Environ. Mutag.* **1**, 87-92.
8. Snedecor, G. W. & Cochran, W. G. (1967) *Statistical Methods* (Iowa State Univ. Press, Ames, IO), 6th Ed.
9. Zeiger, E., Chhabra, R. S. & Margolin, B. H. (1979) *Mutat. Res.* **64**, 379-389.
10. Seiler, J. P., Mattern, I. E., Green, M. H. L. & Anderson, D. (1980) *Mutat. Res.* **74**, 71-75.
11. Gossett, W. S. [Student] (1907) *Biometrika* **5**, 351-360.
12. Fisher, R. A. (1941) *Ann. Eugenics* **11**, 182-187.
13. Haynes, R. H. & Eckardt, F. (1980) in *Chemical Mutagens, Principles and Methods for Their Detection*, eds. Hollaender, A. & de Serres, F. J. (Plenum, New York), Vol. 6, pp. 271-307.
14. Rao, C. R. (1973) *Linear Statistical Inference and Its Applications* (Wiley, New York), 2nd Ed.